DATA STRUCTURE AND PROGRAM DESIGN LAB – 02

2. Design, Develop and Implement a menu driven Program in C for the following operations on STACK of Integers (Array Implementation of Stack with maximum size MAX)

a. Push an Element on to Stack
b. Pop an Element from Stack
c. Demonstrate how Stack can be used to check Palindrome
d. Demonstrate Overflow and Underflow situations on Stack
e. Display the status of Stack
f. Exit

Support the program with appropriate functions for each of the above operations.

SAMPLE OUTPUT:

```
--- STACK MENU ---
1. Push
2. Pop
3. Check Palindrome
4. Display Stack
5. Exit

Enter your choice: 1
Enter value to push: 10

10 pushed to stack.

--- STACK MENU ---
1. Push
2. Pop
3. Check Palindrome
4. Display Stack
5. Exit
```

```
Enter your choice: 1
Enter value to push: 20

20 pushed to stack.

--- STACK MENU ---
1. Push
2. Pop
3. Check Palindrome
4. Display Stack
5. Exit

Enter your choice: 1
Enter value to push: 30

30 pushed to stack.
```

```
--- STACK MENU ---
1. Push
2. Pop
3. Check Palindrome
4. Display Stack
5. Exit

Enter your choice: 4

Stack elements are:
30
20
10

--- STACK MENU ---
1. Push
2. Pop
3. Check Palindrome
4. Display Stack
5. Exit
```

```
Enter your choice: 2

30 popped from the stack.

--- STACK MENU ---
1. Push
2. Pop
3. Check Palindrome
4. Display Stack
5. Exit

Enter your choice: 4

Stack elements are:
20
10
```

```
--- STACK MENU ---
1. Push
2. Pop
3. Check Palindrome
4. Display Stack
5. Exit

Enter your choice: 3

Enter a string: madam

m a d a m is a Palindrome.

--- STACK MENU ---
1. Push
2. Pop
3. Check Palindrome
4. Display Stack
5. Exit
```

```
Enter your choice: 3

Enter a string: hello

hello is Not a Palindrome.

--- STACK MENU ---
1. Push
2. Pop
3. Check Palindrome
4. Display Stack
5. Exit

Enter your choice: 5
```